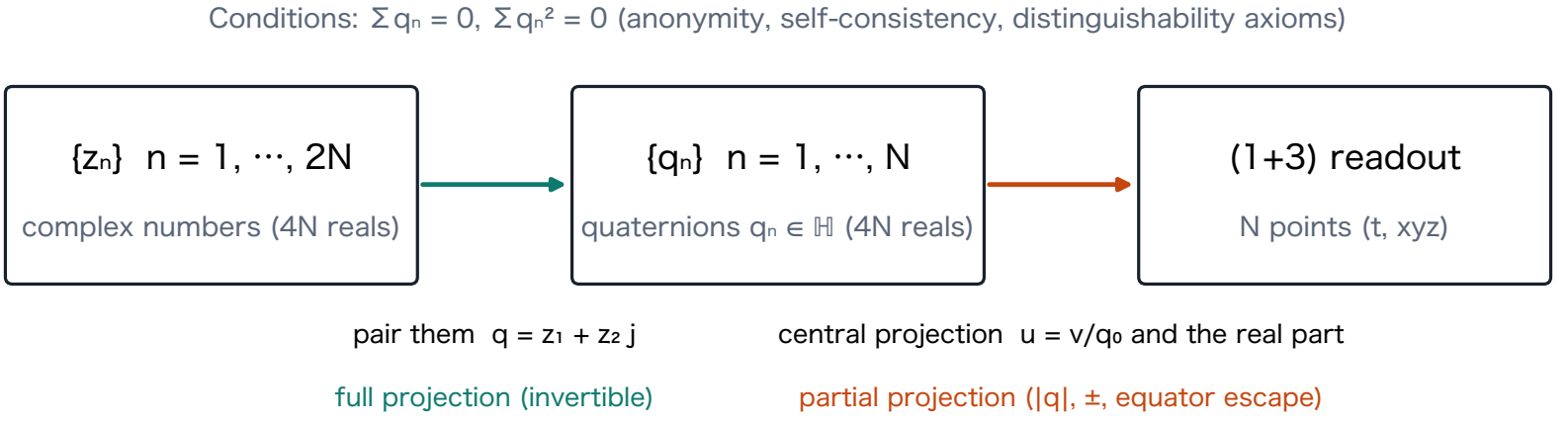
Conditions: $\Sigma q_n = 0$, $\Sigma q_n^2 = 0$ (anonymity, self-consistency, distinguishability axioms)
$\{z_n\}$ n = 1, ⋯, 2N
complex numbers (4N reals)
$\{q_n\}$ n = 1, ⋯, N
quaternions $q_n \in \mathbb{H}$ (4N reals)
(1+3) readout
N points (t, xyz)
pair them $q = z_1 + z_2 j$
full projection (invertible)
central projection $u = v/q_0$ and the real part
partial projection (|q|, ±, equator escape)